

Multiple Linear Regression

1 From SLR to MLR

Recap: simple linear regression

The model with one regressor:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$

OLS estimators, written through the sample covariance, the variances and the correlation coefficient r :

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad (1)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-2}. \quad (2)$$

Question: is one regressor enough?

Delivery time of a soft-drink route driver depends on the number of cases loaded (x_1) and the walking distance (x_2).

Why not simply fit two separate SLRs, y on x_1 and y on x_2 ?

- The two regressors are typically correlated (more cases usually means longer distance).
- Each SLR slope then absorbs part of the effect of the omitted regressor, so it is not the effect of that regressor alone.
- Separate fits cannot answer “what is the effect of one more case for a fixed distance?”.
- The residual variance is also larger because the omitted regressor’s effect is left in the error.

Example: delivery time data

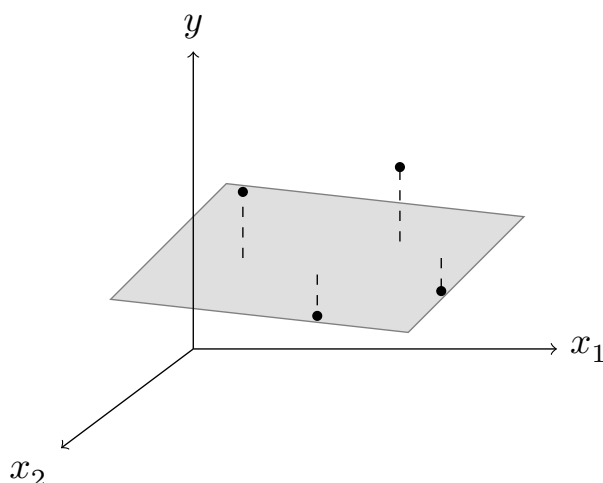
Montgomery, Peck & Vining, Example 3.1 ($n = 25$).

- y : delivery time (minutes)
- x_1 : number of cases stocked
- x_2 : distance walked by the driver (feet)

Obs.	Time y	Cases x_1	Distance x_2
1	16.68	7	560
2	11.50	3	220
3	12.03	3	340
$\vdots$	$\vdots$	$\vdots$	$\vdots$
25	10.75	4	150

From a line to a plane

- SLR: a line in the (x, y) plane.
- Two regressors: a plane in (x_1, x_2, y) space.
- k regressors: a hyperplane in $(k + 1)$ dimensions.
- Residuals: vertical distances to the fitted surface.



2 The MLR model

The MLR equation

With k regressors and n observations:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, \dots, n. \quad (3)$$

- x_{ij} : value of regressor j for observation i .
- $p = k + 1$ unknown regression coefficients $\beta_0, \dots, \beta_k$ (including the intercept).
- ε_i : random error of observation i .

Meaning of the coefficients

$$\frac{\partial E[y \mid x_1, \dots, x_k]}{\partial x_j} = \beta_j, \quad j = 1, \dots, k.$$

- β_j is the change in the mean response per unit increase in x_j when all other regressors are held fixed.
- For this reason $\beta_1, \dots, \beta_k$ are called partial regression coefficients.
- β_0 is the mean response when all regressors are zero (meaningful only if that point is within the range of the data).

Question: interpreting the coefficients

In the delivery time model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$, interpret β_1 , β_2 and β_0 in words, with units.

From n equations to one matrix equation

Writing (3) out for every observation:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \cdots + \beta_k x_{1k} + \varepsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_{21} + \cdots + \beta_k x_{2k} + \varepsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_{n1} + \cdots + \beta_k x_{nk} + \varepsilon_n \end{aligned}$$

Stack the responses, the coefficients and the errors into vectors, and the regressor values into a matrix.

The MLR model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{4}$$

$$\underbrace{\mathbf{y}}_{n \times 1}, \quad \underbrace{\mathbf{X}}_{n \times p}, \quad \underbrace{\boldsymbol{\beta}}_{p \times 1}, \quad \underbrace{\boldsymbol{\varepsilon}}_{n \times 1}.$$

The vectors $\mathbf{y}$, $\boldsymbol{\beta}$, $\boldsymbol{\varepsilon}$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

The design matrix $\mathbf{X}$

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}$$

- Row i is observation i ; column j is regressor j .
- The first column of ones corresponds to the intercept β_0 .

Model assumptions

$$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \tag{5}$$

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n. \tag{6}$$

- $\mathbf{X}$ is fixed (non-random) and has full column rank, i.e. $\text{rank}(\mathbf{X}) = p \leq n$.
- No distributional assumption on $\boldsymbol{\varepsilon}$ is needed yet for estimation (however normality assumption is required later for inference)

$\mathbb{E}[\boldsymbol{\varepsilon}]$ is a n dimensional vector of zeroes.

What does $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n$ say?

$$\text{Var}(\varepsilon) = \text{E}[\varepsilon \varepsilon'] = \text{E}[(\varepsilon - \mathbf{0})(\varepsilon - \mathbf{0})'] = \begin{pmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{pmatrix}$$

- $\text{Var}(\varepsilon)$ is $n \times n$ dimensional variance-covariance matrix.
- Diagonal entries: the i -th diagonal entry contains the variance of the i -th error term. In this model, every error has the same variance σ^2 (constant variance).
- Off-diagonal entries: cell (i, l) contains the covariance between observations i and l . In this model, errors of different observations are uncorrelated, hence covariances are zero.

“Linear” means linear in the parameters

Each of these is an MLR model:

$$\begin{aligned} y &= \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon, \\ y &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon. \end{aligned}$$

This one is not:

$$y = \beta_0 e^{\beta_1 x} + \varepsilon.$$

Polynomial and interaction terms are handled later in the course.

3 OLS estimation in matrix form

The least-squares criterion

Choose β to minimise the sum of squared errors

$$S(\beta) = \sum_{i=1}^n \varepsilon_i^2 = \varepsilon' \varepsilon = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta). \quad (7)$$

Expanding $S(\beta)$

$$\begin{aligned} S(\beta) &= \mathbf{y}'\mathbf{y} - \beta'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta + \beta'\mathbf{X}'\mathbf{X}\beta \\ &= \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta. \end{aligned} \quad (8)$$

The second line combines the two middle terms.

Why is $\mathbf{y}'\mathbf{X}\beta = \beta'\mathbf{X}'\mathbf{y}$?

Two rules for differentiating with respect to a vector

For a constant vector $\mathbf{a}$ and a constant symmetric matrix $\mathbf{A}$:

$$\frac{\partial \mathbf{a}'\beta}{\partial \beta} = \mathbf{a}, \quad \frac{\partial \beta'\mathbf{A}\beta}{\partial \beta} = 2\mathbf{A}\beta.$$

Apply with $\mathbf{a} = \mathbf{X}'\mathbf{y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$ (symmetric) to (8).

The normal equations

$$\begin{aligned}\frac{\partial S}{\partial \beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\beta, \\ \left. \frac{\partial S}{\partial \beta} \right|_{\beta=\hat{\beta}} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{0}, \\ \Rightarrow \mathbf{X}'\mathbf{X}\hat{\beta} &= \mathbf{X}'\mathbf{y}.\end{aligned}\tag{9}$$

This is a system of p linear equations in the p unknowns $\hat{\beta}_0, \dots, \hat{\beta}_k$.

The OLS estimator

If $\mathbf{X}'\mathbf{X}$ is invertible, the normal equations (9) have the unique solution

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.\tag{10}$$

When is $\mathbf{X}'\mathbf{X}$ invertible?

- $\mathbf{X}'\mathbf{X}$ ($p \times p$) is invertible if and only if $\mathbf{X}$ has full column rank.
- That requires $n \geq p$ and no column of $\mathbf{X}$ being a linear combination of the other columns.
- If a regressor is (nearly) a linear combination of the others, $\mathbf{X}'\mathbf{X}$ is singular (nearly singular): this is the collinearity problem, discussed in a later lecture.

Is $\hat{\beta}$ a minimum?

$$\frac{\partial^2 S}{\partial \beta \partial \beta'} = 2\mathbf{X}'\mathbf{X}.$$

- For any $\mathbf{a} \neq \mathbf{0}$: $\mathbf{a}'\mathbf{X}'\mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})'(\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|^2 > 0$, because full column rank gives $\mathbf{X}\mathbf{a} \neq \mathbf{0}$.
- So $\mathbf{X}'\mathbf{X}$ is positive definite and $\hat{\beta}$ is the unique minimiser of (7).

4 Fitted values, residuals and the hat matrix

Fitted values and residuals

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ \hat{\mathbf{y}} &= \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}, \\ \mathbf{H} &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}',\end{aligned}\tag{11}$$

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}.\tag{12}$$

$\mathbf{H}$ is the $n \times n$ hat matrix: it puts the hat on $\mathbf{y}$.

Properties of the hat matrix

$$\mathbf{H}' = \mathbf{H}, \quad (13)$$

$$\mathbf{H}^2 = \mathbf{H}, \quad (14)$$

$$\mathbf{H}\mathbf{X} = \mathbf{X}. \quad (15)$$

That is, $\mathbf{H}$ is symmetric and idempotent.

Prove (14) and (15) from the definition (11).

Properties of $\mathbf{I} - \mathbf{H}$

From (13), (14) and (15):

$$(\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}, \quad (16)$$

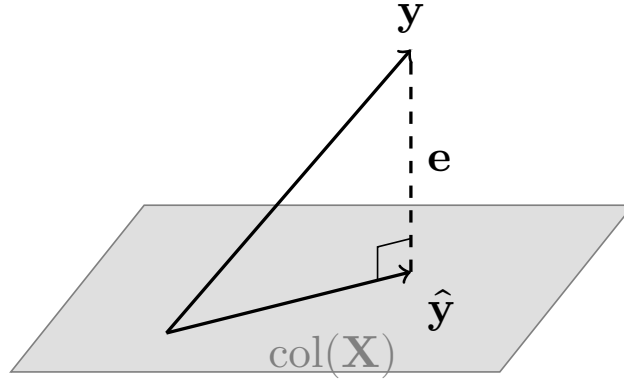
$$(\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - \mathbf{H}, \quad (17)$$

$$(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}. \quad (18)$$

Geometry of least squares

$\mathbf{H}$ is symmetric and idempotent and $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, therefore, $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{X}$.

- The columns of $\mathbf{X}$ span a p -dimensional subspace $\text{col}(\mathbf{X})$ of $\mathbb{R}^n$.
- $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto $\text{col}(\mathbf{X})$.
- $\mathbf{e}$ is perpendicular to $\text{col}(\mathbf{X})$.
- $\hat{\mathbf{y}}$ is the point of $\text{col}(\mathbf{X})$ closest to $\mathbf{y}$, so $\hat{\beta}$ minimises $\|\mathbf{y} - \mathbf{X}\beta\|^2$.



Orthogonality of residuals and regressors

$$\mathbf{X}'\mathbf{e} = \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{0}, \quad (19)$$

by the normal equations (9). Consequences:

- First row of $\mathbf{X}'$ is a row of ones, so $\sum_i e_i = 0$.
- Row j gives $\sum_i x_{ij}e_i = 0$ for each regressor.
- $\hat{\mathbf{y}}'\mathbf{e} = \hat{\beta}'\mathbf{X}'\mathbf{e} = 0$.

5 Delivery time example: OLS fit

Computing $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{y}$

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 25 & 219 & 10\,232 \\ 219 & 3\,055 & 133\,899 \\ 10\,232 & 133\,899 & 6\,725\,688 \end{pmatrix},$$
$$\mathbf{X}'\mathbf{y} = \begin{pmatrix} 559.60 \\ 7\,375.44 \\ 337\,071.69 \end{pmatrix}.$$

The inverse

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 1.13 \times 10^{-1} & -4.45 \times 10^{-3} & -8.37 \times 10^{-5} \\ -4.45 \times 10^{-3} & 2.74 \times 10^{-3} & -4.79 \times 10^{-5} \\ -8.37 \times 10^{-5} & -4.79 \times 10^{-5} & 1.23 \times 10^{-6} \end{pmatrix}$$

The fitted equation

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \begin{pmatrix} 2.3412 \\ 1.6159 \\ 0.01438 \end{pmatrix}$$
$$\hat{y} = 2.3412 + 1.6159 x_1 + 0.01438 x_2.$$

Interpreting the fit

- Each additional case adds about 1.62 minutes to the delivery time, for a fixed distance.
- Each additional foot adds about 0.0144 minutes (about 1.4 minutes per 100 ft), for a fixed number of cases.
- The SLR of y on x_1 alone gives a slope of 2.18, not 1.62: cases and distance are correlated ($r = 0.82$), and the SLR slope absorbs part of the distance effect.

Verifying in R

```
y <- delivery$time
X <- cbind(1, delivery$cases, delivery$distance)

solve(t(X) %*% X, t(X) %*% y)

coef(lm(time ~ cases + distance,
        data = delivery))
```

- `solve(A, b)` solves $\mathbf{Ax} = \mathbf{b}$ directly; it is numerically preferable to forming $\mathbf{A}^{-1}$ explicitly.

6 Properties of the OLS estimators

Linearity

$$\hat{\beta} = \mathbf{A}\mathbf{y}, \quad \mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \quad (20)$$

$\mathbf{A}$ is a $p \times n$ matrix of constants (given $\mathbf{X}$), so every $\hat{\beta}_j$ is a linear combination of $y_1, \dots, y_n$.

Decomposing $\hat{\beta}$

Substitute the model (4) into (10):

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon. \end{aligned} \quad (21)$$

The estimator is the true β plus a linear function of the errors.

Unbiasedness

Taking expectations in (21) with $\mathbf{X}$ fixed, and using (5):

$$\begin{aligned} E[\hat{\beta}] &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\varepsilon] \\ &= \beta. \end{aligned}$$

The covariance matrix of a random vector

$\hat{\beta}$ is p dimensional random vector. Therefore, $\text{Var}(\hat{\beta})$ is a $p \times p$ symmetric matrix:

- diagonal entries: $\text{Var}(\hat{\beta}_j)$;
- off-diagonal entries: $\text{Cov}(\hat{\beta}_i, \hat{\beta}_j)$.

For a constant matrix $\mathbf{B}$ and a random vector $\mathbf{z}$, directly from the definition,

$$\text{Var}(\mathbf{B}\mathbf{z}) = \mathbf{B} \text{Var}(\mathbf{z}) \mathbf{B}'. \quad (22)$$

Covariance matrix of $\hat{\beta}$

$$\begin{aligned} \text{Var}(\hat{\beta}) &= \text{Var}(\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon) \\ &= \text{Var}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{Var}(\varepsilon) \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}. \end{aligned} \quad (23)$$

The steps use (21) (the constant β adds no variance), (22) and (6).

Reading the covariance matrix

Let $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}$, with entries C_{ij} , $i, j = 0, \dots, k$.

$$\begin{aligned}\text{Var}(\hat{\beta}_j) &= \sigma^2 C_{jj}, \\ \text{Cov}(\hat{\beta}_i, \hat{\beta}_j) &= \sigma^2 C_{ij}, \quad i \neq j.\end{aligned}\tag{24}$$

Question: what makes $\text{Var}(\hat{\beta}_j)$ large?

Using (24), which features of the data and the design make the variance of $\hat{\beta}_j$ large?

- Large error variance σ^2 .
- Small sample size n .
- Little spread in regressor $\mathbf{x}_j$ (as in SLR, $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$).
- Strong linear relationship between $\mathbf{x}_j$ and the other regressors: C_{jj} becomes large. This is the collinearity problem.

6.1 Distribution of the estimates under normal errors

Distribution of $\hat{\beta}$

Add the assumption $\varepsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. By (21), $\hat{\beta}$ is a linear function of a normal vector, hence normal:

$$\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}).\tag{25}$$

The mean and covariance matrix are the ones derived above. This result drives the inference procedures in the coming weeks.

7 The Gauss–Markov theorem

Gauss–Markov theorem

Under (5) and (6) (no normality needed), the OLS estimator $\hat{\beta}$ is the best linear unbiased estimator (BLUE) of β .

- Linear: a linear function of $\mathbf{y}$.
- Unbiased: $E[\hat{\beta}] = \beta$.
- Best: among all linear unbiased estimators, each $\hat{\beta}_j$, and more generally each $\mathbf{c}'\hat{\beta}$, has the smallest variance.

8 Maximum Likelihood Estimation of MLR Parameters

Maximum likelihood under normal errors

The likelihood function is given by:

$$\begin{aligned}L(\beta, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i'\beta)^2}{2\sigma^2}\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right).\end{aligned}$$

The log-likelihood of $(\boldsymbol{\beta}, \sigma^2)$ is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \quad (26)$$

For any fixed σ^2 , maximising over $\boldsymbol{\beta}$ is the same as minimising $S(\boldsymbol{\beta})$ in (7), so the ML estimator of $\boldsymbol{\beta}$ equals the OLS estimator.

Maximum likelihood estimator of σ^2

Setting $\partial\ell/\partial\sigma^2 = 0$ at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives

$$\hat{\sigma}_{\text{ML}}^2 = \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n} = \frac{\text{SSE}}{n}. \quad (27)$$

We show next that this estimator is biased.

9 Estimating the variance parameter σ^2

Residuals as a function of the errors

The residual vector can also be written as:

$$\begin{aligned} \mathbf{e} &= \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y} \\ &= (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}, \end{aligned} \quad (28)$$

Equation (28) holds because $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$ as per (18). Then the residual sum of squares:

$$\begin{aligned} \text{SSE} &= \sum_{i=1}^n e_i^2 = \mathbf{e}'\mathbf{e} \\ &= \boldsymbol{\varepsilon}'(\mathbf{I} - \mathbf{H})'(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}. \end{aligned} \quad (29)$$

Equation (29) holds because $(\mathbf{I} - \mathbf{H})$ is symmetric and idempotent as per equations (16) and (17).

Expectation of a quadratic form

For a constant symmetric matrix $\mathbf{B}$:

$$\begin{aligned} \mathbf{E}[\boldsymbol{\varepsilon}'\mathbf{B}\boldsymbol{\varepsilon}] &= \mathbf{E}[\text{tr}(\boldsymbol{\varepsilon}'\mathbf{B}\boldsymbol{\varepsilon})] = \mathbf{E}[\text{tr}(\mathbf{B}\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')] \\ &= \text{tr}(\mathbf{B} \mathbf{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']) = \sigma^2 \text{tr}(\mathbf{B}). \end{aligned} \quad (30)$$

- A scalar equals its trace.
- $\text{tr}(\mathbf{MN}) = \text{tr}(\mathbf{NM})$, and expectation ($\mathbf{E}$) commutes with tr .
- $\mathbf{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \sigma^2\mathbf{I}_n$ because $\mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ as mentioned in (5) and $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ as in (6), therefore, $\mathbf{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}] = \mathbf{E}[(\boldsymbol{\varepsilon} - \mathbf{0})(\boldsymbol{\varepsilon} - \mathbf{0})'] = \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$.

The trace of $\mathbf{I} - \mathbf{H}$

$$\begin{aligned} \text{tr}(\mathbf{H}) &= \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \text{tr}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) \\ &= \text{tr}(\mathbf{I}_p) = p, \\ \text{tr}(\mathbf{I}_n - \mathbf{H}) &= n - p. \end{aligned} \quad (31)$$

$E[\text{SSE}]$ and an unbiased estimator of σ^2

Applying (30) with $\mathbf{B} = \mathbf{I} - \mathbf{H}$ to (29), and using (31):

$$E[\text{SSE}] = \sigma^2(n - p), \quad (32)$$

$$\hat{\sigma}^2 = \text{MSE} = \frac{\text{SSE}}{n - p}. \quad (33)$$

$E[\text{MSE}] = \sigma^2$. For $p = 2$ this is the SLR estimator (2).

Question: why divide by $n - p$?

The n residuals are $e_i = y_i - \hat{y}_i$. Looking at (32), why is the divisor of SSE $n - p$ and not n ? What is the bias of $\hat{\sigma}_{\text{ML}}^2$ in (27)?

- The residuals satisfy the p linear constraints $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (as seen in equation 19), so only $n - p$ of them are free, therefore, $n - p$ degrees of freedom
- $E(\text{SSE}/n) = \sigma^2(n - p)/n$ which is smaller than σ^2
- So the ML estimator of the variance term, $\hat{\sigma}_{\text{ml}}^2 = \text{SSE}/n$ underestimates σ^2

Standard errors of the coefficients

Replace σ^2 by $\hat{\sigma}^2$ in (23) and (24):

$$\begin{aligned} \widehat{\text{Var}}(\hat{\beta}) &= \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}, \\ \text{se}(\hat{\beta}_j) &= \sqrt{\hat{\sigma}^2 C_{jj}}. \end{aligned} \quad (34)$$

Delivery time data: $\hat{\sigma}^2$

- $\text{SSE} = 233.73$, with $n - p = 25 - 3 = 22$ degrees of freedom.
- $\hat{\sigma}^2 = \text{MSE} = 233.73/22 = 10.62$.
- $\hat{\sigma} = 3.26$ minutes: the typical deviation of a delivery time from the fitted plane.

Delivery time data: variance-covariance matrix of the estimates

From (23) with σ^2 replaced by $\hat{\sigma}^2 = 10.62$ and $(\mathbf{X}'\mathbf{X})^{-1}$ from before:

$$\begin{aligned} \widehat{\text{Var}}(\hat{\beta}) &= \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1} \\ &= 10.62 \begin{pmatrix} 1.13 \times 10^{-1} & -4.45 \times 10^{-3} & -8.37 \times 10^{-5} \\ -4.45 \times 10^{-3} & 2.74 \times 10^{-3} & -4.79 \times 10^{-5} \\ -8.37 \times 10^{-5} & -4.79 \times 10^{-5} & 1.23 \times 10^{-6} \end{pmatrix} \\ &\approx \begin{pmatrix} 1.20 & -4.73 \times 10^{-2} & -8.89 \times 10^{-4} \\ -4.73 \times 10^{-2} & 2.91 \times 10^{-2} & -5.09 \times 10^{-4} \\ -8.89 \times 10^{-4} & -5.09 \times 10^{-4} & 1.31 \times 10^{-5} \end{pmatrix}. \end{aligned}$$

- The diagonal entries are $\widehat{\text{Var}}(\hat{\beta}_j)$; their square roots are the standard errors 1.097, 0.171 and 0.00362 in (34).
- The off-diagonal entries are estimated covariances; e.g. $\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2) = -5.09 \times 10^{-4} < 0$: the estimates of the cases and distance coefficients are negatively correlated, because the regressors are positively correlated ($r = 0.82$).

Delivery time data: standard errors

Using (34):

Coefficient	Estimate	Standard error
β_0 (intercept)	2.3412	1.0967
β_1 (cases)	1.6159	0.1707
β_2 (distance)	0.01438	0.003613

10 Coefficient of determination and adjusted R^2

Sums of squares

$$\begin{aligned} \text{SST} &= \sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}'\mathbf{y} - n\bar{y}^2, \\ \text{SSE} &= \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}, \\ \text{SSR} &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - n\bar{y}^2, \\ \text{SST} &= \text{SSR} + \text{SSE}. \end{aligned} \tag{35}$$

The decomposition follows from the orthogonality (19).

The coefficient of determination

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}. \tag{36}$$

- The proportion of the variability in y explained by the regressors; $0 \leq R^2 \leq 1$.
- Delivery time data: $R^2 = 0.9596$.

Adding an irrelevant regressor

What happens to R^2 when a regressor is added to the model? Does it depend on whether the regressor is relevant?

Add z , 25 independent $N(0, 1)$ draws unrelated to the delivery time.

Model	p	R^2	MSE
cases	2	0.9305	17.484
cases, distance	3	0.9596	10.624
cases, distance, z	4	0.9597	11.109

R^2 increases slightly with the useless regressor, while the MSE goes up.

Adjusted R^2

$$R_{\text{adj}}^2 = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)} = 1 - \frac{\text{MSE}}{\text{MST}}, \quad (37)$$

$$= 1 - \frac{n-1}{n-p} (1 - R^2). \quad (38)$$

$\text{MST} = \text{SST}/(n-1)$ is the sample variance of y .

Why adjusted R^2 penalises useless regressors

- MST does not depend on the model, so by (37) R_{adj}^2 increases if and only if MSE decreases.
- Adding a regressor lowers SSE but also lowers the degrees of freedom $n-p$ in (33).
- The MSE decreases only if the drop in SSE outweighs the lost degree of freedom.

Adjusted R^2 in the example

Model	p	R^2	R_{adj}^2
cases	2	0.9305	0.9275
cases, distance	3	0.9596	0.9559
cases, distance, z	4	0.9597	0.9539

R_{adj}^2 rises when the relevant regressor (distance) is added and falls when the irrelevant z is added.

Properties of adjusted R^2

From (38):

- For $p > 1$, $R_{\text{adj}}^2 < R^2$ unless $R^2 = 1$.
- It can decrease when a regressor is added.
- It can be negative: $R_{\text{adj}}^2 < 0$ whenever $R^2 < (p-1)/(n-1)$.

However,

- A high R^2 or R_{adj}^2 does not mean the model is adequate: the assumptions must be checked with residual analysis.
- R_{adj}^2 is a descriptive measure, not a hypothesis test.
- Formal comparison of models is done later with F-tests and information criteria.

11 SLR as a Special Case of MLE

SLR as a special Case of MLR

For SLR, $\mathbf{X} = [\mathbf{1} \ \mathbf{x}]$ and

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}.$$

Since $n \sum x_i^2 - (\sum x_i)^2 = nS_{xx}$,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}.$$

The second entry of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is

$$\hat{\beta}_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{nS_{xx}} = \frac{nS_{xy}}{nS_{xx}} = \frac{S_{xy}}{S_{xx}},$$

which is (1).

SLR variance from the MLR covariance matrix

In equation (23) we found that $\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Using the inverse computed for SLR, with $\sum x_i^2 = S_{xx} + n\bar{x}^2$:

$$\begin{aligned} \text{Var}(\hat{\beta}_1) &= \sigma^2 \frac{n}{nS_{xx}} = \frac{\sigma^2}{S_{xx}}, \\ \text{Var}(\hat{\beta}_0) &= \sigma^2 \frac{\sum x_i^2}{nS_{xx}} = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= -\sigma^2 \frac{\sum x_i}{nS_{xx}} = -\sigma^2 \frac{\bar{x}}{S_{xx}}. \end{aligned}$$

These are the familiar SLR results.

12 Summary

Summary

- Model: $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, with $E[\varepsilon] = \mathbf{0}$ and $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n$ (4).
- OLS: $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ (10); $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a projection onto $\text{col}(\mathbf{X})$.
- $\hat{\beta}$ is unbiased with $\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ (23), and is BLUE.
- $\hat{\sigma}^2 = \text{SSE}/(n-p)$ is unbiased for σ^2 (33).
- R^2 never decreases when regressors are added; R_{adj}^2 (37) penalises the loss of degrees of freedom.

Next lecture

- Inference on the regression coefficients: t -tests and confidence intervals.
- Test for significance of regression (F-test).
- Reading: Montgomery, Peck & Vining, Chapter 3.

A Proof of the Gauss–Markov Theorem

Gauss–Markov: setup of the proof

Any linear estimator can be written

$$\tilde{\beta} = [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}]\mathbf{y}$$

for some $p \times n$ matrix $\mathbf{D}$. Its expectation is $\beta + \mathbf{DX}\beta$, so unbiasedness for every β requires

$$\mathbf{DX} = \mathbf{0}. \tag{39}$$

Gauss–Markov: the variance comparison

$$\begin{aligned} \text{Var}(\tilde{\beta}) &= \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}][\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'] \\ &= \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{DX})' \\ &\quad + \mathbf{DX}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{DD}'] \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{DD}', \end{aligned}$$

where the last step uses (39).

Gauss–Markov: conclusion

- $\mathbf{a}'\mathbf{DD}'\mathbf{a} = \|\mathbf{D}'\mathbf{a}\|^2 \geq 0$ for every $\mathbf{a}$, so $\mathbf{DD}'$ is positive semidefinite.
- Hence $\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta})$ is positive semidefinite, and $\text{Var}(\mathbf{c}'\tilde{\beta}) \geq \text{Var}(\mathbf{c}'\hat{\beta})$ for every $\mathbf{c}$.
- Equality holds only for $\mathbf{D} = \mathbf{0}$, i.e. $\tilde{\beta} = \hat{\beta}$.